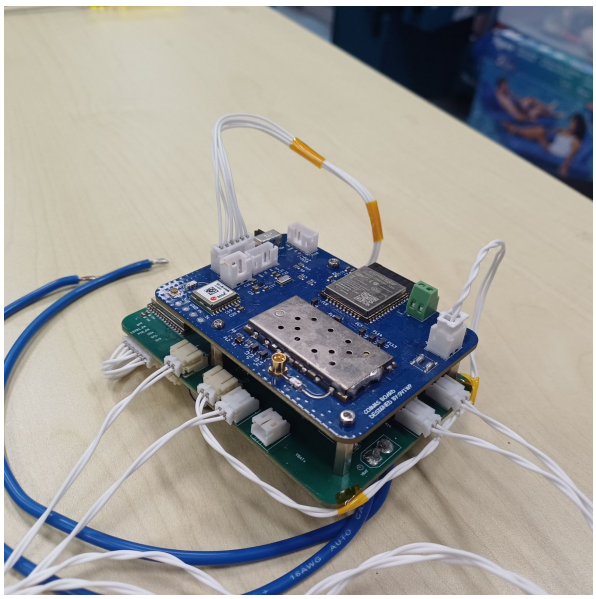
Development and Flight Testing of a Student-Developed Avionics Stack for STAR High-Altitude Balloon Payloads

Power Distribution, Telemetry and RF Subsystems flown on BEAR-13 (*Project Strato TV*) and BEAR-14 (*NUS TEH HALIA-1*)

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Abstract

This report documents the design, ground verification and flight testing of a modular avionics stack developed by undergraduate students for high-altitude balloon (HAB) payloads flown under the NUS Satellite Technology and Research Centre (STAR). The stack comprises (i) a four-layer *communications board* integrating a 144.39 MHz APRS transmitter, a multi-constellation GNSS receiver, a high-range barometer and controlled-impedance RF routing, and (ii) a four-layer *power distribution board* (PDB) providing a regulated 5 V buck–boost rail, five individually switched payload channels, battery-rail current sensing and closed-loop battery heating. The communications board flew on BEAR-13 (29 March 2026), which carried Southeast Asia's first live analogue video downlink from a weather balloon at 1.28 GHz; the full stack flew on BEAR-14 (30 August 2026). Analysis of the archived APRS telemetry shows that both flights exceeded 31 km (peak 33.5 km), with 94–97 % frame recovery through the terrestrial APRS network. On BEAR-14 the PDB maintained the battery between 15.1 °C and 32.8 °C while the avionics bay reached 11.8 °C, at a mean system draw of 3.2 W and a consumption of under 30 % of the measured pack energy. A recomputed link budget for the BEAR-13 video downlink predicts carrier-to-noise ratio falling through the FM threshold at a slant range of ≈46 km, matching the observed point of total video loss and correcting an 11 dB free-space-loss error in the pre-flight estimate. The analysis also identifies several defects to be resolved before the next flight: Wi-Fi-induced GNSS desensitisation, a cutter state machine that re-arms instead of latching, telemetry-side barometric quantisation of up to 570 m, and sensor starvation during blocking SSTV transmission.

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1 Introduction

1.1 Background

Latex weather balloons offer low-cost access to the stratosphere and are widely used as a training platform for spacecraft engineering: payloads must survive ambient temperatures below $-50\,^{\circ}\mathrm{C}$, pressures under 10 hPa, rapid thermal transients and multi-hour unattended operation, while meeting radio-frequency (RF) and aviation regulatory constraints. In Singapore, the Balloon Experiments with Amateur Radio (BEAR) group has flown amateur payloads since before the start of this work. NUS TEH HALIA is the first student cohort of a STAR balloon programme that uses these flights to develop flight hardware in-house.

Earlier BEAR payloads were assembled from development boards and point-to-point wiring (Figure 1). Such harnesses are difficult to inspect, offer no telemetry on power-system health, and provide no means of switching payloads off during the long pad holds that are common on launch day. The objective of the work reported here was to replace this arrangement with a stack of purpose-designed printed circuit boards that can be re-used across STAR payloads.

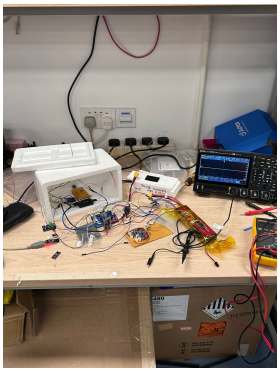


Figure 1. Pre-programme bench harness for an earlier BEAR payload, built from development boards and point-to-point wiring. The avionics stack described in this report was developed to replace this arrangement.

1.2 Scope and contributions

This report covers two boards, their firmware and two flights:

1. **Communications (telemetry) board**: APRS/SSTV transmitter, GNSS, barometer, flight-termination (cutter) logic and camera control. Flown on BEAR-13 and BEAR-14.
2. **UREx power distribution board**: wide-input 5 V conversion, switched payload rails, current sensing and battery heating. Flown on BEAR-14.
3. **Flight data analysis**: a post-flight analysis of the archived APRS telemetry from both flights, performed for this report, covering trajectory, power, thermal, barometric and link performance.

Hardware and firmware sources are held in the public repositories `RandomAerospace/COMMS-BOARD`, `RandomAerospace/power_stuff` and `RandomAerospace/BEAR13_FIRMWARE` [1, 2, 3]. The tracking antenna rotator (N. Bong) and the BEAR-13 battery management board (A. Tan) are described only where they interact with the stack.

1.3 Flight summary

Table 1. Flights covered by this report. Times are UTC; Singapore local time is UTC+8.

	BEAR-13 (*Project Strato TV*)	**BEAR-14** (*NUS TEH HALIA-1*)
Date (launch, UTC)	28 Mar 2026, 22:51	30 Aug 2026, 00:15
Callsign	9V1WP-11	9V1WP-11
Primary payload	1.28 GHz analogue FPV video, two cameras	SSTV on 145.670 MHz, STAR student payload
Avionics flown	Comms board + battery management board (A. Tan) + drone PDB	Comms board + UREx PDB
Battery	3S Li-Po, 5200 mA h	2S pack, 5228 mA h measured
Peak GNSS altitude	31 093 m	33 504 m
APRS frames recovered	96.9 % (370/382)	93.9 % (246/262)

2 Requirements and System Architecture

2.1 Mission concept of operations

The BEAR-13 concept of operations (ConOps) set the baseline for the stack. The payload transmits APRS position and housekeeping telemetry on the amateur VHF APRS frequency of 144.390 MHz every 30 s. Frames are relayed by the terrestrial APRS digipeater and internet-gateway (iGate) network, which extends tracking beyond the line of sight of any single ground station [4]. In parallel, a 1.5 W analogue video transmitter (VTx) at 1280 MHz alternates between a horizon and a zenith camera on every APRS cycle and is received by a tracking antenna. The rotator points the antenna using positions retrieved from `aprs.fi`, driving a repurposed telescope mount through Hamlib. The flight-termination set point is 30 km; the zenith camera is selected for five minutes before the cut. BEAR-14 replaced the video downlink with SSTV image transmission on 145.670 MHz, sent immediately after an APRS frame on the same radio, and added the PDB and a STAR student payload.

Flights were cleared with the Republic of Singapore Air Force (203 Squadron), the Civil Aviation Authority of Singapore and the Infocomm Media Development Authority (IMDA). IMDA approval was required for the wideband 1.28 GHz emission.

2.2 Design requirements

Table 2 lists the PDB requirements as defined at the start of the BEAR-14 development cycle. The verification column is completed from the evidence presented in Sections 6 and 7.

Table 2. Power distribution board requirements and verification status.

ID	Requirement	Implementation	Status
R1	Deliver 5 V@2 A each to a Raspberry Pi, an SDR and one APRS board (≈6 A peak); allow for a future digital video payload	TPS552892 buck–boost (WEBENCH design, 5 A); three TPS22997 load switches	Partial: 15 W load on one channel
R2	Current sensing and bus voltage monitoring	INA232 with 6.8 mΩ shunt on battery rail	Verified in flight
R3	Enable/disable payloads for power cycling and saving	TCA6408A I/O expander driving all switched rails; Wi-Fi control page	Verified on pad
R4	Battery heating	PTC heater on a TPS1HB08 high-side switch; DS18B20 feedback	Verified in flight
R5	Accept 5–18 V input	Buck–boost input range; 16.8 V (4S) design maximum	Not characterised

2.3 Stack architecture

Figure 2 shows the BEAR-14 configuration. The PDB and the communications board each carry an ESP32 module. The PDB microcontroller (the *head*) owns the power system: it samples the INA232 and the battery temperature sensor at 1 Hz, runs the heater controller, drives the rail switches through the I/O expander, and pushes a 13-byte packed telemetry structure to the communications board (the *tail*) over I^2C at address `0x42`. The tail owns the radio and the navigation sensors, and multiplexes the received power telemetry into alternate APRS frames. Splitting the microcontrollers this way means that a fault in the radio firmware cannot leave the power rails or the heater in an undefined state, and the PDB can be reused under a different telemetry system.

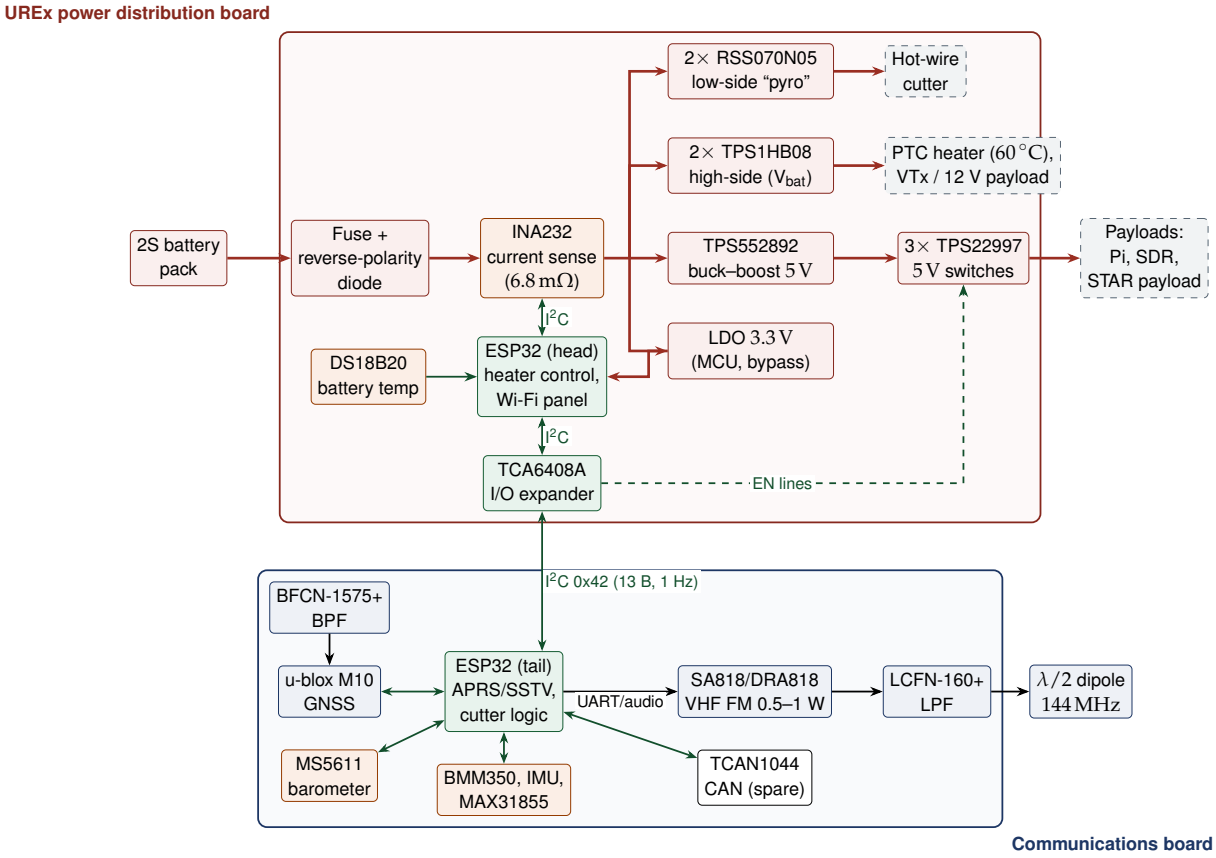


Figure 2. BEAR-14 avionics architecture. Thick red arrows carry power, thin arrows carry signals. Dashed boxes are external loads. On the PDB the head ESP32 is powered from a dedicated LDO on the battery rail, so the controller stays alive if the buck–boost converter fails.

3 Communications Board

3.1 Overview

The communications board (75 × 55 mm, four layers, 107 footprints, 426 vias) is built around an ESP32-WROOM-32E module. It carries an SA818-family (DRA818V-compatible) VHF FM transceiver module, a u-blox M10 GNSS receiver, an MS5611-01BA barometer, a BMM350 magnetometer with an IMU, a MAX31855 type-K thermocouple interface reserved for an external probe, and a TCAN1044 CAN transceiver reserved for future stack expansion. Selected schematic sheets are reproduced in Appendix A.

3.2 RF front ends and impedance control

Two RF paths share the board with a 1.5 W, 1280 MHz video transmitter mounted centimetres away. Coexistence was the main layout driver:

- **GNSS receive path.** A Mini-Circuits BFCN-1575+ ceramic band-pass filter sits ahead of the passive patch antenna input to reject the 1.28 GHz VTx carrier.
- **VHF transmit path.** A Mini-Circuits LCFN-160+ low-pass filter at the transceiver output suppresses harmonics of the 144.39 MHz carrier.
- **Controlled impedance.** Both paths are routed as coplanar waveguide with ground (CPWG) on the outer layer, referenced to an unbroken inner ground on L2, with solder mask removed over the RF traces and a via fence along both sides.

- **Low-frequency filtering.** Data lines to the radio and sensors pass through 220 Ω @ 100 MHz ferrite beads (BLM21PG221) and common-mode chokes, which present a low-pass response below 100 MHz. This keeps VHF transmit energy from coupling into the logic supply and digital lines. GPIO-only lines (PTT, power, squelch) are left unfiltered by design.

The fabricator's JLC7628 stack-up (0.2104 mm 7628 prepreg between L1 and L2) was entered into the vendor's impedance solver, giving a 0.342 mm trace with a 0.5 mm gap for 50 Ω (Figure 3a). To check this independently, the conformal-mapping closed form for CPWG [6, 7] was evaluated with a thickness correction for 35 µm copper:

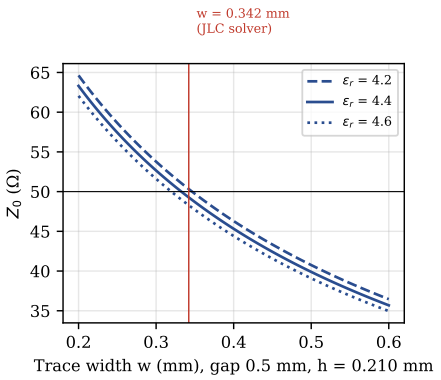
$$Z_0 = \frac{60\pi}{\sqrt{\varepsilon_{\text{eff}}}} \left[\frac{K(k)}{K(k')} + \frac{K(k_1)}{K(k_1')} \right]^{-1}, \qquad k = \frac{w}{w+2s}, \quad k_1 = \frac{\tanh(\pi w/4h)}{\tanh(\pi(w+2s)/4h)}, \tag{1}$$

where $K(\cdot)$ is the complete elliptic integral of the first kind and $k' = \sqrt{1-k^2}$. For $\varepsilon_r = 4.2 - -4.6$ the trace gives $Z_0 = 48.3 - -50.3\,\Omega$ (Figure 3b), within the $\pm 10\,\%$ fabrication tolerance of the solver result. The effective permittivity of 3.36 gives a guided wavelength of 128 mm at 1280 MHz. The design rule of stitching vias at $\lambda/8$ of the VTx frequency should therefore be applied to the guided wavelength, giving a pitch of $\leq$16 mm, not the 29 mm implied by the free-space value. At the GNSS L1 frequency the corresponding figure is 13 mm.

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Width	Trace Spacing	Impedance trace to copper
50	Coplanar Single Ended	L1	/	L2	0.3421	/	0.5000

Layer	Material	Thickness (mil)	Thickness (mm)
L1	Outer Copper Weight1oz	1.38	0.0350
Prepreg	7628, RC 49%, 8.6 mil	8.28	0.2104
L2	Inner Copper Weight	0.60	0.0152
Core	1.1mm H/HOZ with copper	41.93	1.0650
L3	Inner Copper Weight	0.60	0.0152
Prepreg	7628, RC 49%, 8.6 mil	8.28	0.2104
L4	Outer Copper Weight1oz	1.38	0.0350

(a) Vendor stack-up and CPWG solution.



(b) Independent check using Equation (1).

Figure 3. Controlled-impedance design of the RF traces on the communications board.

3.3 Antenna

The VHF antenna is a half-wave dipole built from coaxial cable and enamelled copper wire and potted with hot-melt adhesive into the payload. A vector network analyser measurement through an MMCX–SMA adaptor gave VSWR $\approx$ 1.67, which corresponds to a return loss of 12 dB and a mismatch loss of 0.52 dB. Two practical lessons were recorded during construction: the enamel must be burned off completely to make a reliable solder joint, and the coax shield should be crimped rather than only soldered, for strain relief.

3.4 Fabrication and inspection

The board was assembled by JLCPCB. X-ray inspection, provided free for boards with ball-grid-array parts, was used to confirm solder joints on the GNSS module. The images also showed the internal construction of the pressure sensor and the GNSS module, including wire bonds on the barometer die. One schematic error reached hardware: an input-only ESP32 pin was assigned to an output function and had to be bodge-wired. The schematic records the as-built reassignment (VTx enable on GPIO5, cutter on GPIO4).

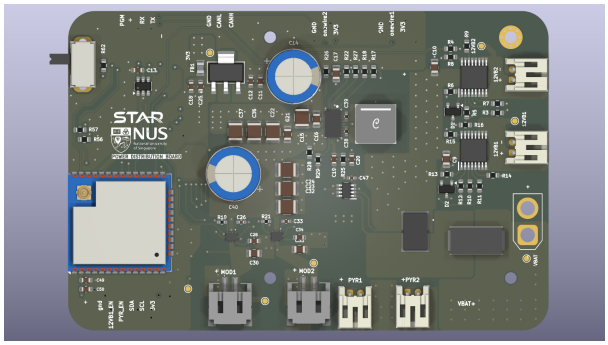
4 Power Distribution Board

4.1 Power tree

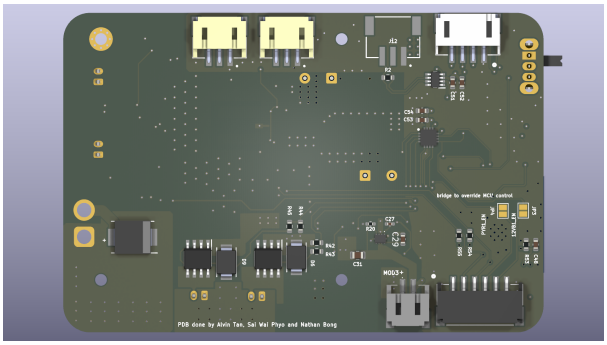
The UREx PDB (89.8 × 59.7 mm, four layers, 128 footprints, 370 vias; Figure 4) implements the power tree of Figure 2. Battery input enters through an XT30 connector, a fuse and a series Schottky diode for reverse-polarity protection. It then passes the 6.8 mΩ current-sense shunt and distributes to four branches:

1. a TPS552892 four-switch buck–boost converter producing 5 V, designed in TI WEBENCH for up to 5 A, with a 4.7 µH (13.6 A saturation) XAL7070 inductor and polymer bulk capacitance (330 µF, 16 mΩ ESR), feeding three TPS22997 load switches;
2. two TPS1HB08-Q1 smart high-side switches on the battery rail, intended for inductive loads such as the VTx, motors and the battery heater, with built-in current limit and diagnostics;
3. two RSS070N05 N-channel MOSFET low-side "pyro" channels for resistive loads such as a hot-wire cutter;
4. a wide-input 3.3 V LDO powering the ESP32 directly from the battery. This bypasses the switching converter, so the controller remains available to command and report even if the converter fails.

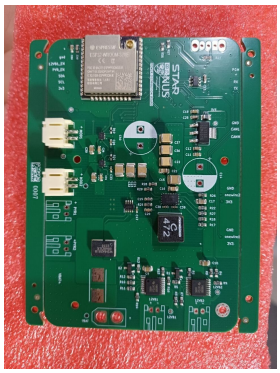
Layer 2 is a split power plane carrying the battery, 5 V and 3.3 V nets (Figure 5). The high-current loop is kept to the top layer between the shunt, converter and output switches.



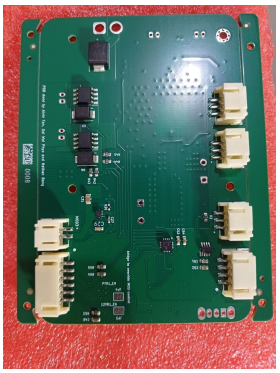
(a) 3D render, top.



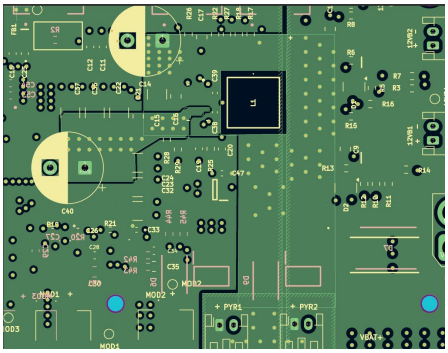
(b) 3D render, bottom.



(c) Assembled, top.



(d) Assembled, bottom.



(e) Layer 2 split power plane.

Figure 4. UREx power distribution board.

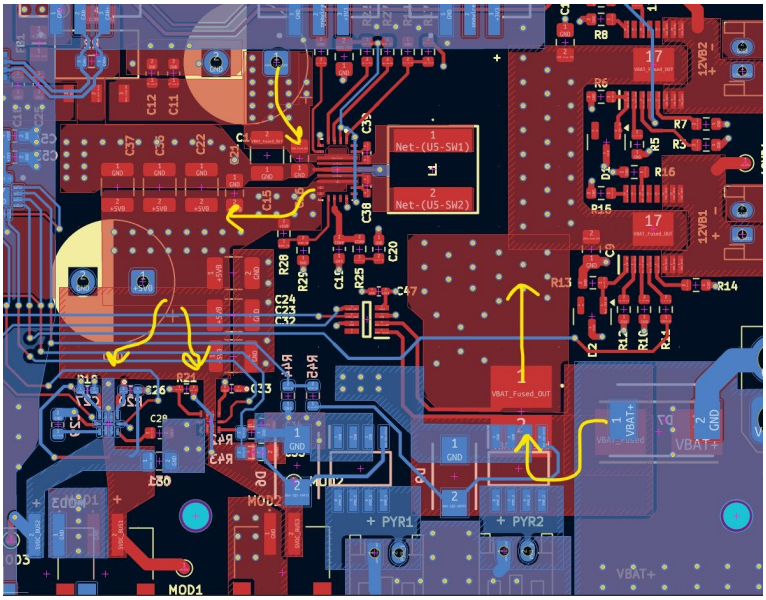


Figure 5. Annotated layout of the power path: fuse, shunt, TPS552892 converter, the three TPS22997 5 V switches, the TPS1HB08 high-side switches, the low-side pyro MOSFETs and the LM1117-class LDO.

4.2 Current sensing

The shunt was sized so that the full-scale input of the INA232 (± 81.92 mV at ADCRANGE = 0) corresponds to the combined current limit of all load switches, $I_{\max} = 12$ A:

$$R_{\text{sh}} \leq \frac{V_{\text{sense,FS}}}{I_{\max}} = \frac{81.92\,\text{mV}}{12\,\text{A}} = 6.83\,\text{m}\Omega, \qquad P_{\text{sh}} = I_{\max}^2 R_{\text{sh}} = 0.98\,\text{W}. \tag{2}$$

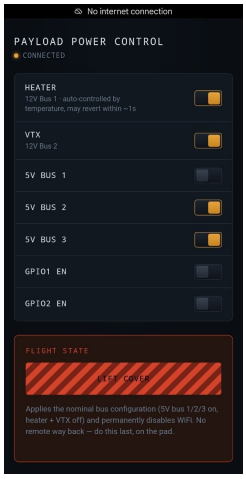
A 6.8 mΩ WSHM2818 power-metal-strip resistor was selected. With the 2.5 µV shunt LSB, the resolution is 0.37 mA. The firmware averages 128 conversions of 8.2 ms, so each reported reading spans about 1 s.

The firmware drives the device through a third-party INA library that assumes the INA230 register map. The two parts share a shunt LSB but differ in bus-voltage LSB (1.25 mV for the INA230, 1.6 mV for the INA232), so the firmware applies a correction factor of $1.6/1.25 = 1.28$ to the bus voltage. This works, but it ties correctness to a constant buried in application code. Some source comments also still refer to an INA228. Both should be cleaned up before the board is reused.

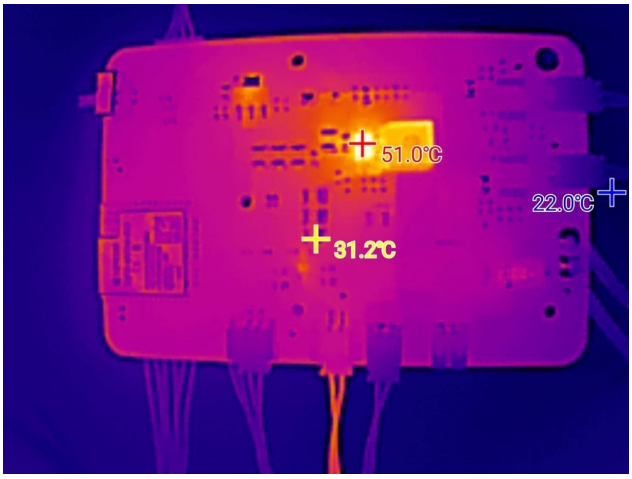
4.3 Load switching and ground control

All seven controllable rails are driven from a TCA6408A I^{2}C I/O expander: three 5 V buses, the heater, the VTx and two general-purpose channels. The firmware keeps a shadow copy of the output register so that automatic heater control and manual commands can change single bits without overwriting each other. In the nominal flight state (`0x38`) the three 5 V buses are on and the heater and VTx are off.

On the pad, the head ESP32 hosts a Wi-Fi access point with a control page (Figure 6a) that toggles each rail and reports live bus status. A single *Enter Flight State* action locks the rail configuration and turns the radio off permanently for the rest of the flight. During BEAR-14 a 90 min launch hold was absorbed by switching payloads off from this page, which validated R3 under operational conditions.



(a) Rail control page.



(b) Thermal image under a 15 W constant-power load at 7.2 V input.

Figure 6. PDB ground control interface and load test.

4.4 Battery pack and thermal control

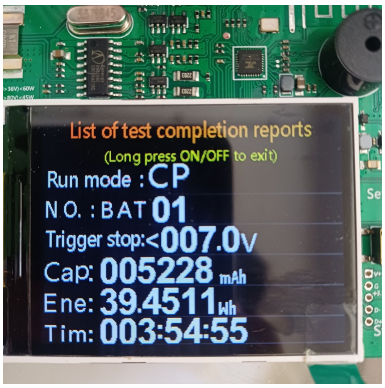
The flight pack was wrapped in polyimide tape with a DS18B20 one-wire temperature sensor and a 60 °C self-limiting PTC heater bonded to the cells (Figure 7). The heater controller is a bang-bang loop with hysteresis. It switches the heater on at $T_{\text{bat}} \leq 15\,^{\circ}\text{C}$ and off at $T_{\text{bat}} \geq 20\,^{\circ}\text{C}$, and forces the heater off if the sensor reports the DS18B20 disconnected value ($-127\,^{\circ}\text{C}$), so that a sensor fault cannot drive the heater on continuously. During integration the dedicated battery one-wire bus gave unreliable readings, so for BEAR-14 the sensor was moved to the second (external) one-wire port.



(a) DS18B20 placement.



(b) PTC heater.



(c) Final conditioning cycle: 5228 mA h, 39.45 W h.

Figure 7. Flight battery pack construction and pre-flight conditioning.

5 Flight Firmware

5.1 Execution model

Both microcontrollers use a cooperative super-loop with `millis()`-based software timers and no real-time operating system. On the communications board, GNSS is polled at 1 Hz, the barometer and temperatures every 2 s, and the cutter altitude check every 1.5 s. Transmissions are sequenced by the state machine in Figure 8. The TRANSMITTING state is blocking. It sends one APRS frame of about 1 s and, when an SSTV image is scheduled, retunes to 145.670 MHz and transmits a full frame of about 36 s before returning. Sensor updates are gated to the IDLE state, so that heavy transmit current does not coincide with sensor reads and GPIO switching. The firmware

comments note that heater current drawn from primary lithium cells could otherwise brown out the ESP32.

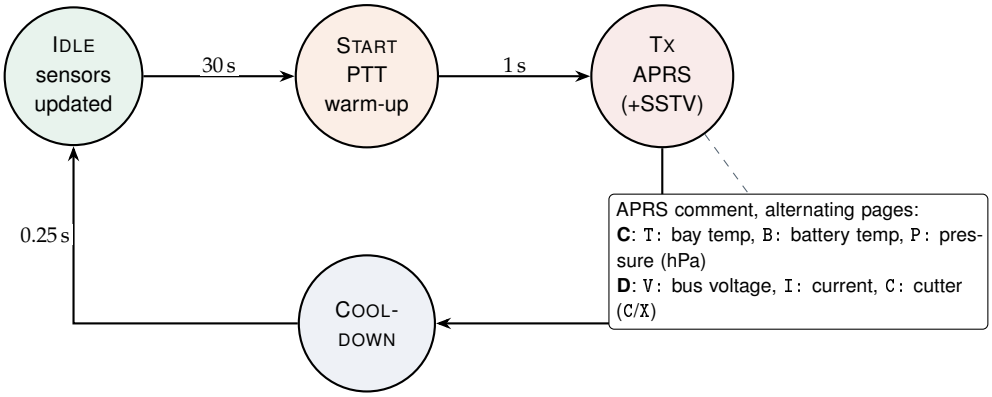


Figure 8. Transmit state machine on the communications board, and the two alternating telemetry pages carried in the APRS comment field.

5.2 Telemetry format

Each APRS frame carries a six-digit frame counter followed by `TxC` or `TxD`, then one of the two housekeeping pages shown in Figure 8. Because the counter increments on every transmission, the gaps between received counters give an exact count of lost frames (Section 7.6). On BEAR-13 the second field on page C was reserved for an external thermocouple (`E:`) and read zero. On BEAR-14 it carried battery temperature from the PDB.

5.3 Inter-board link

The power telemetry link initially ran in the opposite direction, with the comms board as I^2C master. This was reversed during integration, because the PDB's second I^2C peripheral was already committed to the INA232 as master. In the final arrangement the PDB writes a packed structure $\{V_{\mathrm{bus}}, I, T_{\mathrm{bat}}$, heater state$\}$ once per second, and the comms board receives it in an interrupt callback. The structure definitions on the two boards must match byte for byte. Any change to the telemetry fields therefore has to be released to both boards together, which favours a shared header file over two hand-maintained copies.

5.4 GNSS assistance and Wi-Fi

At power-on the comms board joins a local Wi-Fi network, downloads u-blox AssistNow data and seeds the receiver with an approximate position. Wi-Fi is then shut down. A safeguard disables Wi-Fi if the board reboots above 1 km. Two consequences were found in the data. First, the first BEAR-13 fix in the archive (1.352 17° N, 103.819 83° E) is exactly the hard-coded assistance seed, not a measured position. The post-flight spreadsheet used this point as its line-of-sight origin, which distorts its range and elevation columns (Section 7.7). Second, the PDB's own Wi-Fi access point, running at the ESP32 default of 20 dBm, prevented GNSS lock on the pad (Section 8).

5.5 Flight termination logic

The cutter routine declares a cut when GNSS altitude exceeds 30 000 m on five consecutive 1.5 s checks, then holds the cutter output for 15 s. On BEAR-14 the output drive was disabled in firmware, so the logic flew as a dry run with only the telemetry flag active.

6 Ground Testing

6.1 PDB load test

The PDB was loaded at 15 W constant power (3 A at 5 V) from a 7.2 V supply through a single TPS22997 channel. This is representative of a Raspberry Pi 5 under load. The thermal image in Figure 6b shows a hotspot of about 51 °C against a 22 °C background, with a second marked point at 31 °C. The board then powered a Raspberry Pi 5 through one load switch without brown-out. Requirement R1 is therefore only partly verified. The full 5 A converter rating, simultaneous loading of all three channels, and operation at the 5 V and 16.8 V ends of the input range were not tested. Converter efficiency was not measured.

6.2 Battery conditioning

Before flight the pack was taken through three cycles of: constant-power discharge to the cut-off voltage, a 30 min rest, a full recharge, and a second 30 min rest. The final discharge delivered 5228 mA h and 39.45 W h (Figure 7c), an average discharge voltage of 7.55 V, consistent with a healthy 2S pack. This figure is the energy budget used in Section 7.3.

6.3 RF bench test of the video link

For BEAR-13, the VTx output was measured on an RF power meter through 120 dB of fixed attenuation (2×30 dB and 3×20 dB in series), supplied from both 3S (11.1 V) and 2S (7.4 V) sources. The output dropped by about 2 dB on the 2S supply, indicating that the transmitter was undervolted. The VTx antenna was a right-hand circularly polarised (RHCP) design. This choice followed earlier SSTV flights, in which payload roll produced periodic null lines in the received images (Figure 9a), attributed to polarisation mismatch with a linearly polarised antenna.



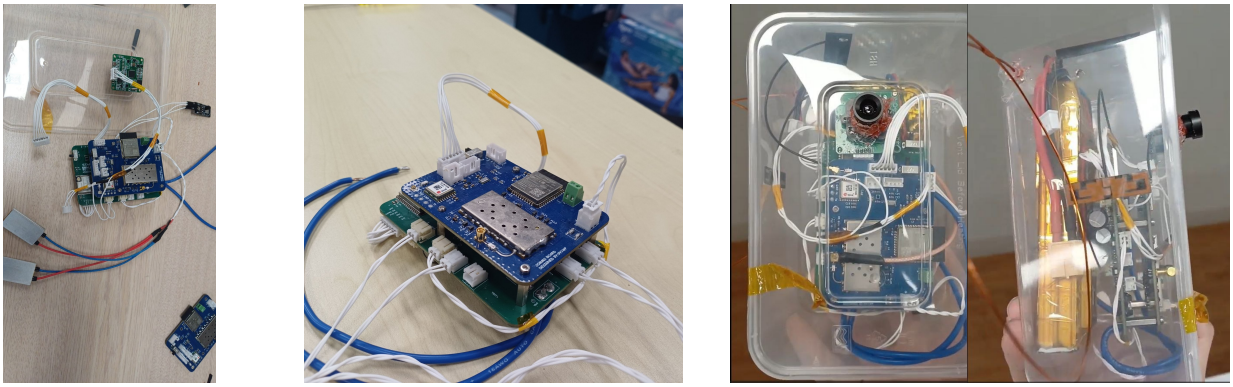
(a) Null lines in a received image from an earlier flight.

(b) Ground-station video monitor, BEAR-13.

Figure 9. Received imagery used to motivate the antenna polarisation choice.

6.4 Integration

Figure 10 shows the BEAR-14 stack during bench integration and in its flight enclosure. The camera intended for this flight was not flown.



(a) Bench bring-up. **(b)** Assembled stack. **(c)** Enclosure. **(d)** Side view.

Figure 10. BEAR-14 avionics integration.

7 Flight Results and Analysis

7.1 Data and method

The flight data are `aprs.fi` exports of every frame received from 9V1WP-11 by the APRS network [3]. BEAR-14 timestamps have one-second resolution. The BEAR-13 export carries only hours and minutes, so BEAR-13 rates and durations are uncertain by about one minute. Positions and altitudes come from the GNSS receiver. Launch is taken as the last sample below 150 m before sustained climb. Rates are least-squares fits over the stated altitude bands. Slant range and elevation are computed on a spherical Earth from the launch site at NUS (1.297 17° N, 103.761 67° E), where the ground station was assumed to be co-located. Pressure altitude is computed with the 1976 US Standard Atmosphere [5]. All analysis scripts are deterministic, and every number in this section was regenerated from the raw exports.

7.2 Trajectory

Table 3. Trajectory parameters derived from the telemetry.

Parameter	BEAR-13	BEAR-14
Launch (UTC)	22:51	00:15
Peak altitude (UTC)	31 093 m (02:36)	33 504 m (02:43)
Time to peak	225 min	148 min
Ascent rate, fitted (1 km to peak)	2.58 m/s ($R^2 = 0.997$)	3.71 m/s ($R^2 = 0.999$)
Descent rate, upper (>15 km)	47.6 m/s	32.7 m/s
Descent rate, lower (0.6–5 km)	15.8 m/s	7.4 m/s
Peak ground speed (altitude)	106 km/h (23.7 km)	152 km/h (14.2 km)
Launch-to-last-fix distance	18.9 km	28.9 km
Maximum slant range from launch site	57.7 km	55.3 km

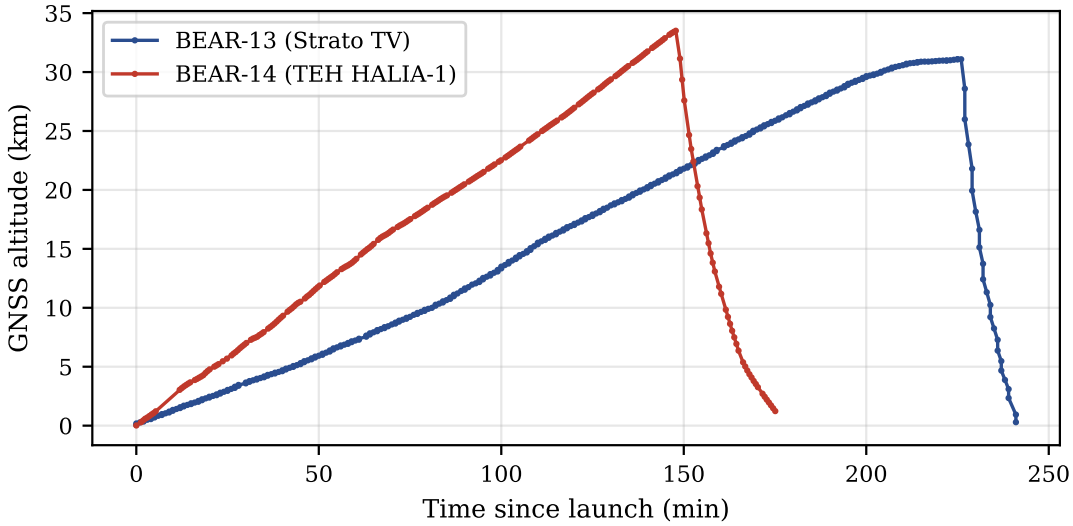
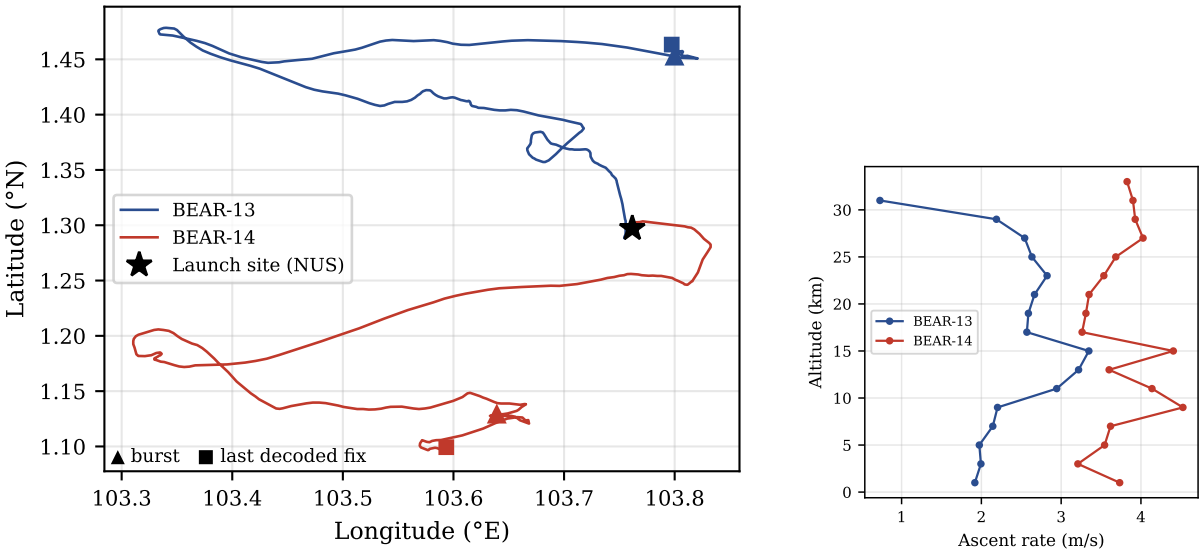


Figure 11. GNSS altitude against time from launch for both flights.

Both ascents are close to linear (Figure 11), as expected for a balloon whose drag and buoyancy scale together with altitude. The lower BEAR-13 rate reflects a smaller free lift. Binned ascent rates (Figure 12b) show BEAR-13 slowing from about 2.5 m/s to 0.7 m/s in its last 2 km, consistent with the balloon approaching its float ceiling. The BEAR-13 cutter flag remained `X` throughout. The telemetry therefore cannot show whether the termination at 31.1 km was a commanded cut or a natural burst. BEAR-14 continued at a steady rate to a natural burst at 33.5 km.

The initial descent speeds of 33–48 m/s are expected, since a parachute generates little drag in the low-density air above 15 km. The near-ground descent rate of 15.8 m/s on BEAR-13, however, is about twice that of BEAR-14 and well above a typical 5–8 m/s landing target. It suggests a parachute that did not open fully or was fouled. Since the descent system is outside the scope of this report, this is flagged for the recovery team rather than diagnosed.

Both payloads drifted west and north-west through strong upper-tropospheric winds, peaking at 152 km/h at 14 km on BEAR-14 (Figure 12a). The last fixes were 19–29 km from the launch site, above Johor and the southern Singapore Strait respectively.



(a) Ground tracks.

(b) Ascent rate, 2 km bins.

Figure 12. Horizontal trajectory and vertical performance.

7.3 Power system performance

Figure 13 shows bus voltage, current and altitude for both flights. Table 4 gives the integrated quantities. Energy and charge are trapezoidal integrals of $V \cdot I$ and I over every power frame, including the pre-launch hold. Because the current telemetry is sampled roughly once every 70 s (every second frame), short events such as transmit bursts are undersampled. The integrals are therefore estimates of average consumption rather than exact totals.

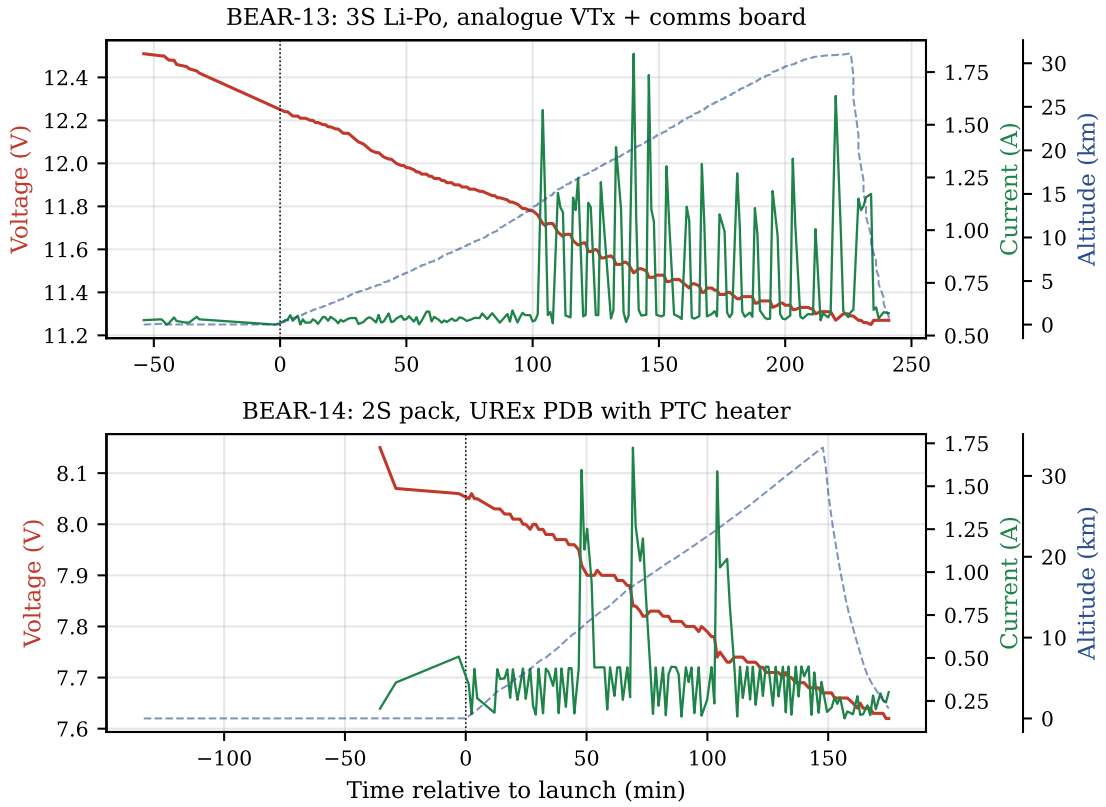


Figure 13. Power telemetry. Current spikes after launch are heater activations: from the BEAR-13 battery management board (threshold 15 °C) and from the PDB heater channel on BEAR-14.

Table 4. Power system figures derived from telemetry.

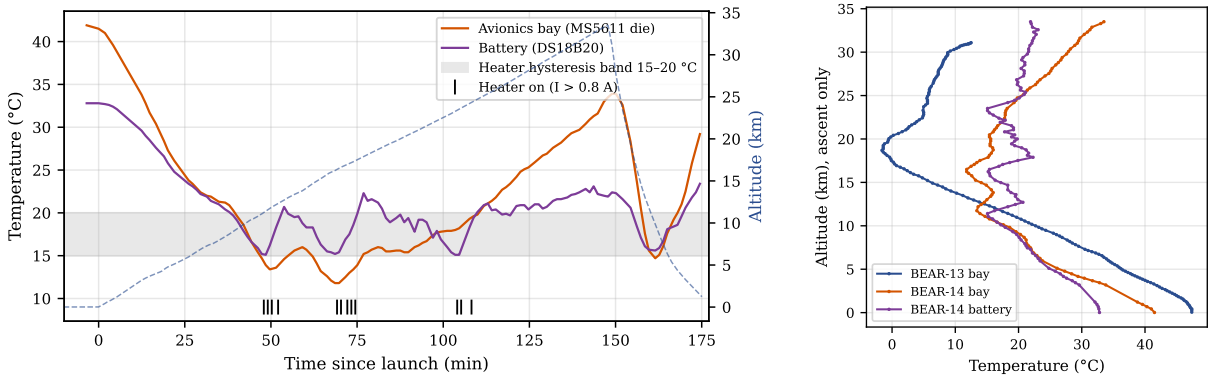
Parameter	BEAR-13 (3S)	BEAR-14 (2S, PDB)
Bus voltage at launch → last frame	12.25 → 11.27 V	8.05 → 7.62 V
Per-cell voltage at last frame	3.76 V	3.81 V
Current, median / mean / max	0.59 / 0.72 / 1.84 A	0.38 / 0.42 / 1.73 A
Logged span	4.9 h	3.5 h
Charge drawn (logged span)	3.41 A h	1.44 A h
Energy drawn (logged span)	40.1 W h	11.3 W h
Mean power	8.1 W	3.2 W
Fraction of pack capacity used	66 % of 5.2 A h	28 % of 5.23 A h

On BEAR-13, the 1.5 W VTx and two cameras dominate the 8.1 W mean load, and two-thirds of the pack was consumed in 4.9 h. The margin was adequate for this flight, but a longer hold or slower ascent would have exhausted the battery before landing. This is the main argument for pad-side power switching (R3), which BEAR-14 provided. On BEAR-14, without the video payload, the stack averaged 3.2 W and used under 30 % of the conditioned pack energy. The bus

voltage sagged only 0.43 V over the flight. The buck–boost converter regulated throughout, with no brown-out or reset visible in the continuous frame counter.

Splitting the BEAR-14 current samples at 0.8 A gives a mean of 1.23 A with the heater on and 0.33 A with it off. The heater therefore draws about 0.90 A, or 7.1 W at 7.9 V. It was on in 9.8 % of power frames, an average heating power of ≈0.7 W.

7.4 Thermal performance



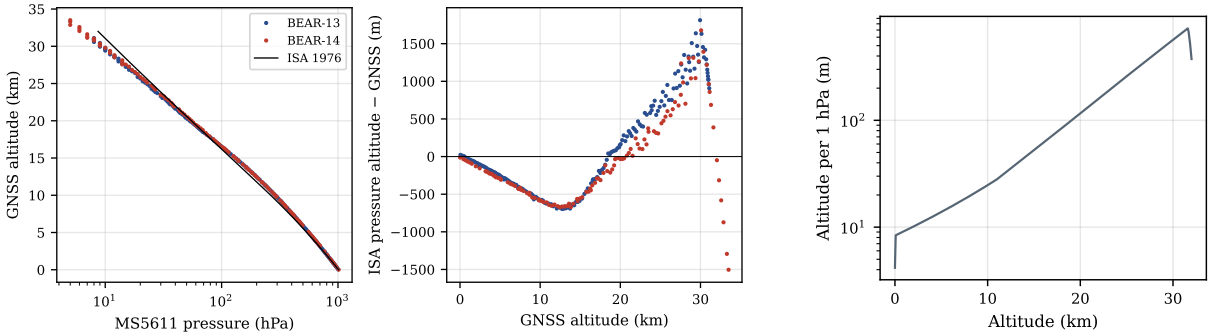
(a) BEAR-14 temperatures and heater activity.

(b) Temperature against altitude, ascent.

Figure 14. Thermal environment inside the payload enclosures.

On BEAR-14 the battery temperature fell from 33 °C at launch into the 15–20 °C control band at about 9 km. It stayed there for the rest of the ascent, with a minimum of 15.1 °C (Figure 14a). Three groups of heater activations near 50, 70 and 105 min line up with recoveries in battery temperature, showing that the loop closed as designed. The avionics-bay temperature, read from the MS5611 die, reached a minimum of 11.8 °C at 16.2 km, close to the tropical tropopause. It then recovered to 34 °C near burst as solar heating increased in the thin stratosphere. The BEAR-13 bay fell to −1.6 °C at 18.7 km (Figure 14b), about 18 K colder than BEAR-14 (16.0 °C) at the same altitude. Some of this difference comes from enclosure design and launch time (06:51 against 08:15 local), so it cannot all be credited to the PDB heater. The battery temperature on BEAR-14, however, was held above the lower limit of the control band through the coldest part of the flight. That is the quantity requirement R4 specifies, and R4 is therefore verified.

7.5 Barometric altitude



(a) Measured pressure and pressure-altitude error.

(b) Altitude step per 1 hPa.

Figure 15. Barometer performance against GNSS altitude.

The MS5611 tracks the standard atmosphere over three orders of magnitude of pressure (Figure 15a). The residual against GNSS has a characteristic shape that repeats on both flights. Below the tropopause the ISA-derived altitude reads low by 250–450 m (mean bias −248 m and −305 m for 0–10 km). Above 20 km it reads high, by up to 1.7 km. Two separate effects explain this:

1. **Atmospheric, not instrumental.** The tropical troposphere is warmer than the ISA reference, so pressure falls off more slowly with height and the ISA pressure altitude underestimates true height. The colder-than-standard tropical tropopause and lower stratosphere then reverse the sign. A barometer calibrated against ISA cannot remove this bias without a local sounding.
2. **Telemetry quantisation.** Pressure is sent as an integer in hectopascals (`%04u`). One hectopascal corresponds to 4 m of altitude at sea level but 570 m at 30 km (Figure 15b). This produces the scatter above 25 km. Above about 31 km the pressure also falls below the MS5611's specified 10 mbar lower limit [8], and the readings of 5–8 hPa near the BEAR-14 burst should be treated as out of specification.

Sending pressure in units of 0.1 Pa, or as the raw sensor counts, would remove effect (2) at no cost in frame length. The pre-flight expectation of usable barometric altitude to 25 km is borne out, subject to the atmospheric bias described above.

7.6 Telemetry link

Frame counters show that the APRS network recovered 96.9 % of BEAR-13 frames and 93.9 % of BEAR-14 frames between launch and the last fix (Figure 16). On BEAR-13 no more than one consecutive frame was lost. The largest BEAR-14 outage was eight frames (about 4.5 min) between 1.2 km and 3.0 km altitude, shortly after launch. The counter continued without a reset across this gap, so the payload kept transmitting and the frames were lost on the receive side. This is consistent with poor low-altitude coverage by iGates near the launch site. One further two-frame gap occurred at 23.7 km. The BEAR-14 radio was configured for the module's low-power (0.5 W) setting, and this performance suggests there is little to gain from the high-power setting given its current cost.

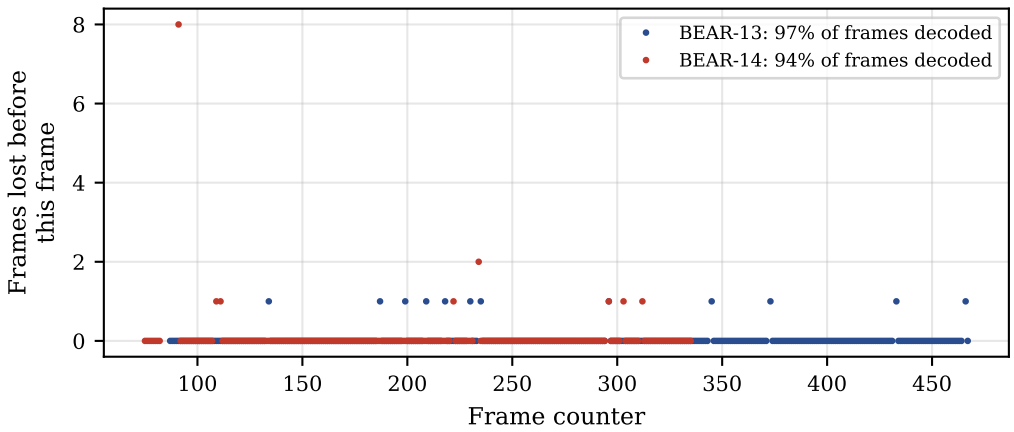


Figure 16. Lost frames preceding each received frame, from gaps in the on-board frame counter.

BEAR-14 also downlinked SSTV images with a telemetry overlay. The frame in Figure 17 was transmitted at 02:40 UTC from 32 874 m, three minutes before burst.



Figure 17. SSTV frame received from BEAR-14 at 32.9 km, with on-board time, position, altitude, speed and pressure overlaid by the communications board.

7.7 BEAR-13 video downlink

The BEAR-13 analogue video degraded at a reported slant range of about 39 km and was lost completely at about 46 km. Recomputing the geometry from the launch site (Figure 18), the payload crossed 39 km slant range at 22.6 km altitude and 35° elevation, and 46 km at 23.6 km altitude and 31° elevation, at 01:32 UTC. The time of the 46 km crossing agrees with the original post-flight spreadsheet. The spreadsheet's elevation of about 60° at that point cannot be reproduced. A payload 23.6 km high and 46 km away must be below 31° elevation. The discrepancy is consistent with the spreadsheet using the AssistNow seed position as its origin, as noted above.

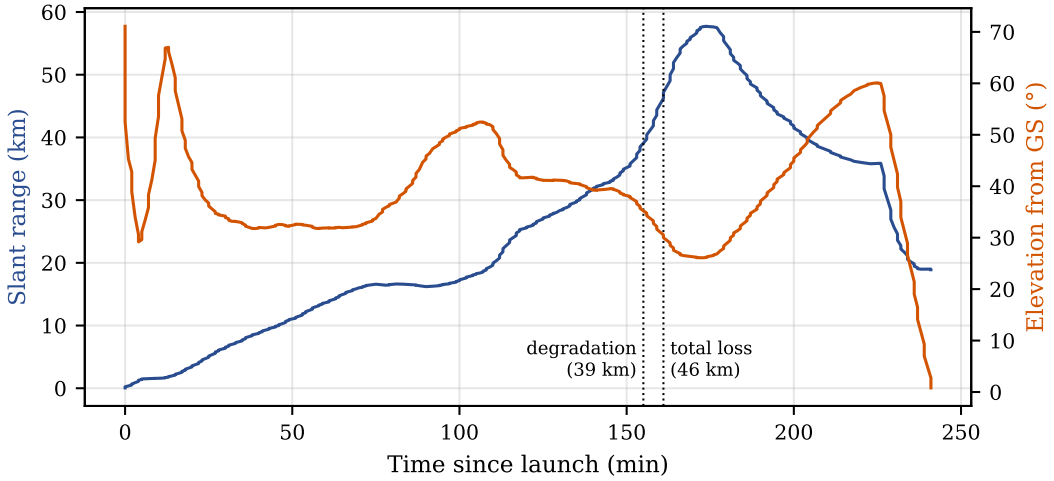


Figure 18. BEAR-13 slant range and elevation from the launch site, with the reported video degradation and loss points.

The pre-flight link estimate used a free-space path loss of 117 dB at 50 km. The Friis free-space loss at 1280 MHz is

$$L_{\mathrm{fs}} = 20\log_{10} d_{\mathrm{km}} + 20\log_{10} f_{\mathrm{MHz}} + 32.44 = 128.6\,\mathrm{dB} \quad (d = 50\,\mathrm{km}), \tag{3}$$

so the estimate was 11.6 dB optimistic. The estimate also treated the 20 MHz FM video signal like a narrowband carrier. A corrected budget with $P_t = 31.8$ dBm (1.5 W), $G_t = 1$ dBi, $G_r = 10$ dBi and 3 dB of miscellaneous cable and pointing loss gives

$$P_r(46\,\mathrm{km}) = 31.8 + 1 + 10 - 127.8 - 3 = -88.1\,\mathrm{dBm}. \tag{4}$$

Against a noise floor of $-174 + 10\log_{10}(20\,\text{MHz}) + NF = -97.0\,\text{dBm}$ for an assumed 4 dB receiver noise figure, this is a carrier-to-noise ratio of 8.9 dB. That is at the FM demodulator threshold of about 8–10 dB, below which analogue video breaks up abruptly. At 39 km the predicted CNR is 10.3 dB, just above threshold, which matches the reported onset of degradation (Figure 19).

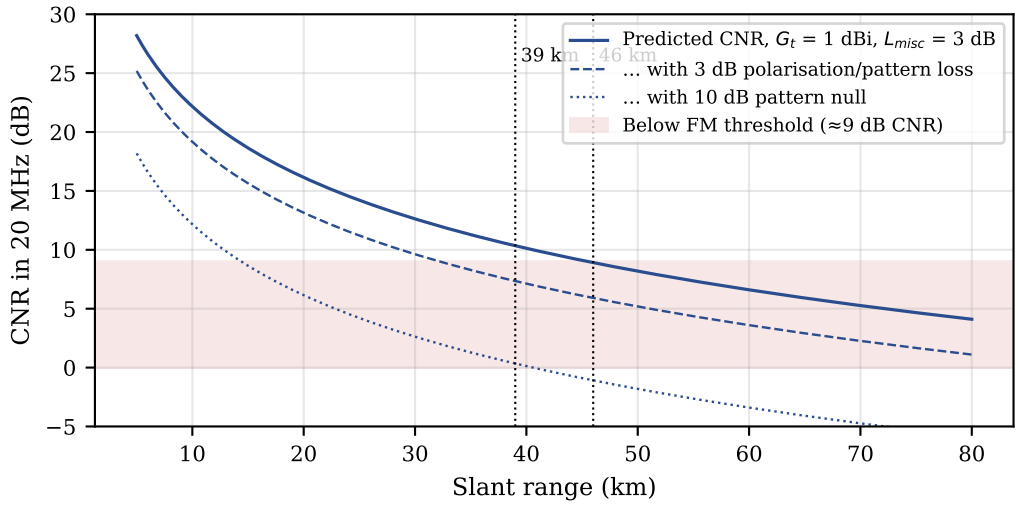


Figure 19. Predicted carrier-to-noise ratio of the 1.28 GHz analogue video link against slant range.

The agreement shows that the video was lost because the link margin ran out, and that no failure mechanism beyond what is in the budget is needed to explain it. The assumed losses are themselves uncertain. A payload antenna 3–10 dB down off boresight at 30° elevation, or RHCP-to-linear polarisation loss, would move the threshold crossing to 15–33 km. That the observed loss point fits the 3 dB case suggests that the RHCP payload antenna kept pattern and polarisation losses small. The suspected interference from the rotator motor drivers was not measured, so neither its contribution nor its absence can be established. For a repeat flight the options, in order of leverage, are:

1. a higher-gain receive antenna. A 20 dBi dish or array adds 10 dB, which roughly triples the usable range.
2. a narrower occupied bandwidth. Digital video in about 2 MHz, as flown by the Horus DATV payloads [13], lowers the noise floor by 10 dB and adds forward error correction gain.
3. more transmit power, which costs battery energy that the BEAR-13 budget (Table 4) could not easily spare.

8 Anomalies and Lessons Learned

Table 5. Anomalies identified from flight, ground test and code review.

#	Anomaly	Evidence / cause	Corrective action
A1	PDB Wi-Fi prevents GNSS lock on the pad	Observed on BEAR-14 at the default 20 dBm, despite the BFCN-1575+ filter. The likely mechanism is front-end compression by a strong 2.4 GHz signal from an antenna a few centimetres away, which the filter cannot reject enough of, rather than in-band emission.	Reduce Tx power (`WiFi.setTxPower`); move ground control to BLE; separate the antennas; enter flight state before GNSS acquisition.
A2	Cutter re-arms instead of latching	On BEAR-14 `C:C` was reported in all 11 power frames above 30.2 km. The altitude check re-asserts the cut on each pass while altitude stays above the set point, and the 15 s cut timer is reset by the blocking transmitter. A live cutter would have fired repeatedly.	Make the cut a one-shot latched event with a "fired" flag; add a mission timer and barometric backup; keep the Tx-inhibit guard.
A3	Sensors starved during SSTV	Sensors and the cutter are not serviced during the blocking 36 s SSTV transmission.	Generate SSTV audio by DMA (I2S/DAC) or move to FreeRTOS tasks.
A4	Barometric quantisation	Integer hectopascals give a 570 m step at 30 km; MS5611 out of specification below 10 hPa.	Transmit 0.1-Pa units or raw counts.
A5	Video link budget	FSPL underestimated by 11.6 dB; the wideband signal was treated as narrowband.	Use the budget in Section 7.7; add receive gain.
A6	Seeded position in data	The first BEAR-13 fix is the AssistNow seed and was used as the spreadsheet origin.	Discard fixes without a 3D solution; log the fix type in telemetry.
A7	Board and code errata	Input-only pin routed to an output function (bodged); battery one-wire port unreliable; INA228/INA230/INA232 inconsistency in code; single `0x38` flight mode.	Next board revision; shared telemetry header; explicit mode state machine.
A8	Fast BEAR-13 landing	15.8 m/s below 5 km, against 7.4 m/s on BEAR-14.	Refer to the recovery team.

Item A2 is the most safety-relevant. A flight-termination system is part of the conditions under which the flights are cleared with aviation authorities. A cutter that cycles with a period set by the radio duty cycle could, with the output enabled, draw repeated pulses from the pyro channel and inhibit telemetry while it does so. The defect only became visible because BEAR-14 deliberately flew the logic as a dry run with telemetry visibility. That was the correct approach, and it should be repeated after the fix and before the output is re-enabled.

9 Conclusions and Future Work

A two-board avionics stack was designed, built and flown by undergraduate students on two STAR/BEAR stratospheric balloon flights, reaching 31.1 km and 33.5 km. The communications board delivered 94–97 % telemetry frame recovery on both flights with independently verified 50 Ω RF routing. On BEAR-13 it coexisted with a 1.5 W 1.28 GHz video transmitter without loss of GNSS. On its first flight the power distribution board regulated the payload bus, reported battery voltage and current throughout, held the battery at or above 15 °C through the tropopause, and allowed payloads to be switched off during a 90 min pad hold. It used under 30 % of the measured pack energy.

The flight data also show where the design needs more work. The video-link analysis showed

that the loss of picture on BEAR-13 is fully explained by link margin once the free-space loss is computed correctly. The cutter dry run exposed a latching defect that would not have been found on the bench. For the next iteration, in priority order:

1. fix the cutter logic (A2) and repeat the dry run before enabling the output;
2. remove the Wi-Fi/GNSS conflict (A1), preferably by moving pad control to BLE;
3. move the communications firmware to a task-based RTOS with non-blocking SSTV (A3);
4. finish characterising the PDB: efficiency over the 5–16.8 V input range, three-channel full load, and the TPS1HB08 current-limit behaviour;
5. use the spare CAN interface on both boards so that additional payload boards can join the stack without new point-to-point wiring;
6. for video, a higher-gain receive antenna or a narrowband digital downlink (DVB-S2 on an SDR), sized with the corrected link budget.

Acknowledgements

The authors thank the NUS Satellite Technology and Research Centre for programme support under UREx; the members of BEAR (9V1FH, 9V1LW, 9V1CV, 9V1YP) for launch, rigging, ground support and regulatory coordination; Bumblebee Autonomous Systems NUS; the Republic of Singapore Air Force 203 Squadron, CAAS and IMDA for flight and spectrum approvals; and 9V1LH, whose BEAR-4 APRS firmware formed the basis of the telemetry code.

References

[1] S. W. Phyo, *COMMS-BOARD: APRS + GNSS board with hot-wire and FPV camera control*, GitHub, `https://github.com/RandomAerospace/COMMS-BOARD`, 2026.

[2] S. W. Phyo, A. Tan, N. Bong, *power_stuff: UREx power distribution board*, GitHub, `https://github.com/RandomAerospace/power_stuff`, 2026.

[3] S. W. Phyo, *BEAR13_FIRMWARE: HAB telemetry and power firmware, with flight data*, GitHub, `https://github.com/RandomAerospace/BEAR13_FIRMWARE`, 2026.

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[7] B. C. Wadell, *Transmission Line Design Handbook*, Artech House, 1991.

[8] TE Connectivity, *MS5611-01BA03 Barometric Pressure Sensor, with stainless steel cap*, datasheet.

[9] Texas Instruments, *TPS552892 Wide-Input Buck-Boost Converter*, datasheet.

[10] Texas Instruments, *INA232 High- or Low-Side, Bidirectional Current and Power Monitor*, datasheet.

[11] Texas Instruments, *TPS22997 Single-Channel Load Switch*, datasheet.

[12] Texas Instruments, *TPS1HB08-Q1 Automotive Single-Channel Smart High-Side Switch*, datasheet.

[13] Amateur Radio Experimenters Group, *Horus DATV*, `https://www.areg.org.au/archives/category/horus-datv`.

A Selected Schematic Sheets

Full schematics are in the repositories [1, 2]. The sheets below are reproduced as vector graphics and can be enlarged in the PDF.

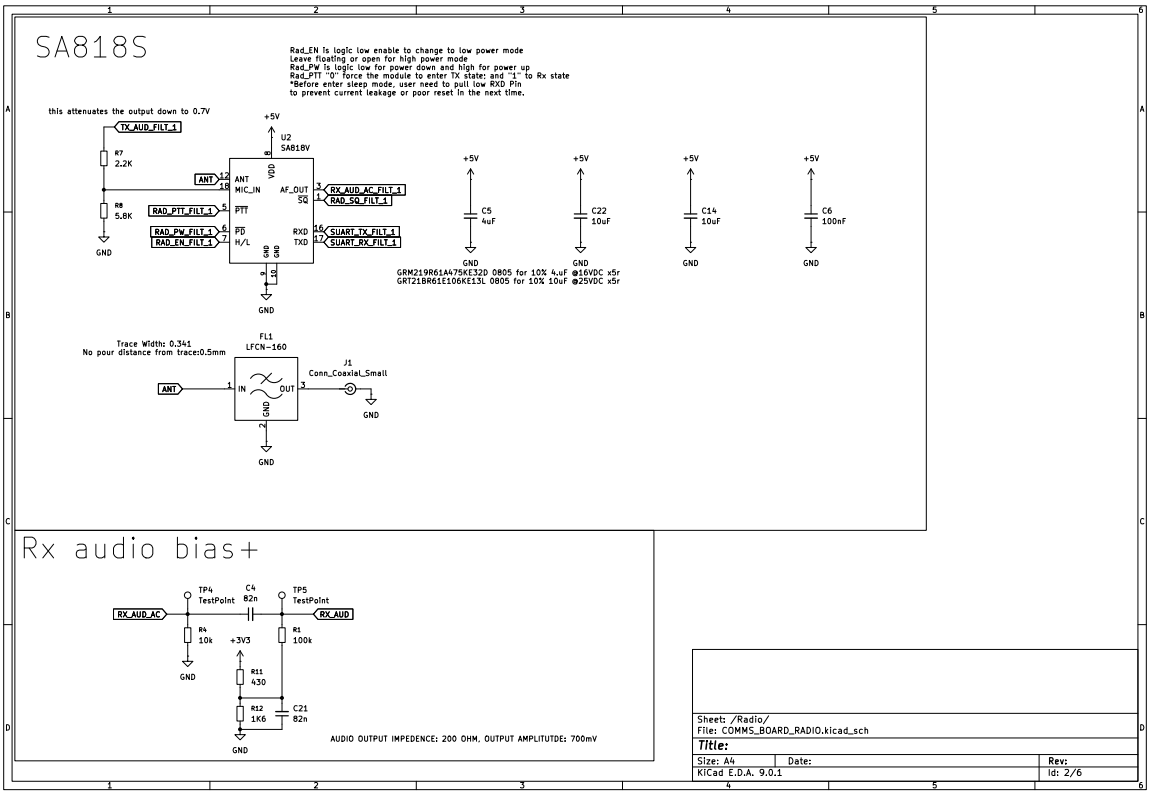


Figure 20. Communications board: SA818 VHF transceiver, output filter and receive-audio bias network.

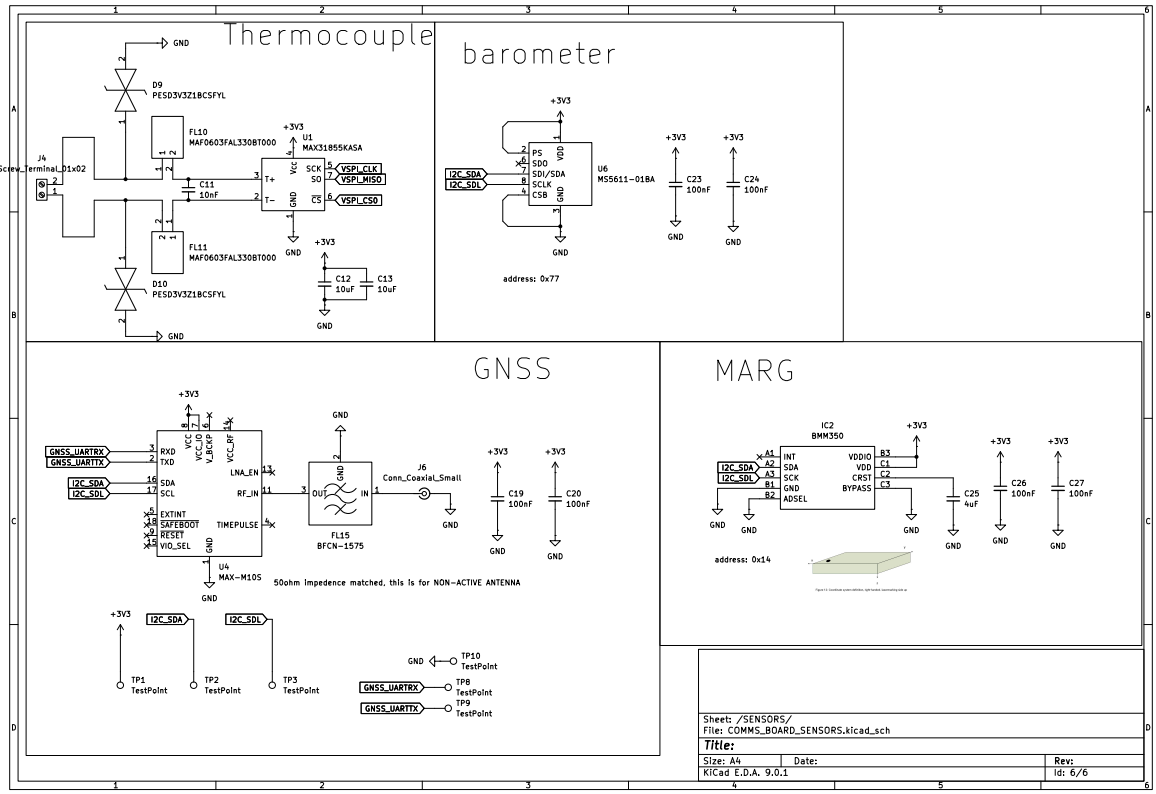


Figure 21. Communications board: GNSS with band-pass filter and 50 Ω feed, MS5611 barometer, thermocouple interface and MARG sensors.

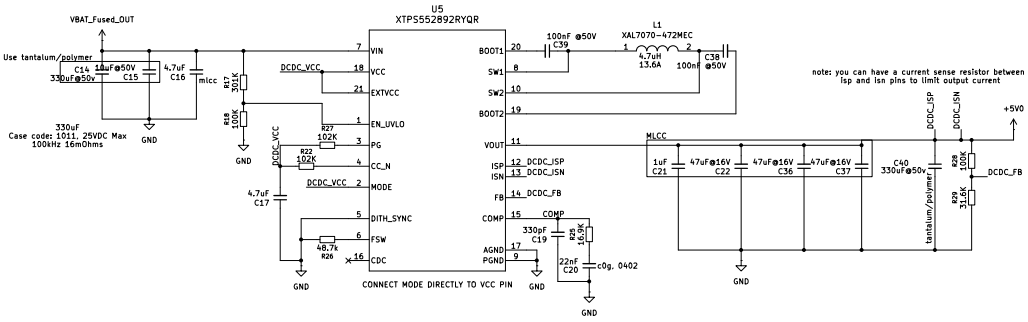


Figure 22. PDB: TPS552892 buck–boost converter.

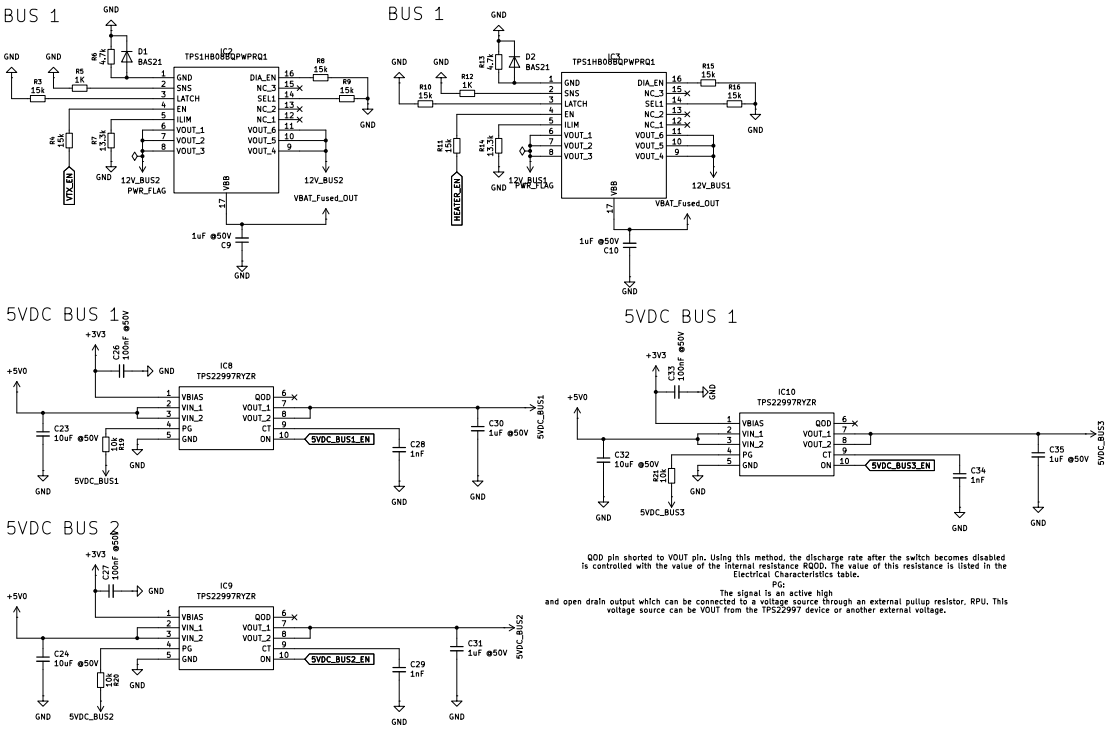


Figure 23. PDB: TPS1HB08 high-side switches and TPS22997 5 V load switches.

B Reproducibility

The figures and statistics in Section 7 were produced by `analyse.py` (trajectory, power, thermal, barometric, packet and link analysis) and `cpwg.py` (Equation (1)), which are supplied with this report. The scripts read the `aprs.fi` exports in `BEAR13_FIRMWARE/FLIGHT DATA/` without manual editing, apart from parsing the minute-resolution BEAR-13 timestamps. Board statistics were extracted from the KiCad `.kicad_pcb` files and production BOMs.